StockEvent Lens — Batch 024 Intelligence Dashboard

Articles 116–120 Analysis, Vector Embedding Sensitivity & Conflict Thresholds

Batch ID
`batch_24 (Articles 116–120)`

TEST 116
Original — 100% Base
IonQ deploys Superion 256 QPU at NVIDIA NVAQC in Boston via NVQLink & CUDA-Q[cite: 37, 38].

TEST 117
Repeat — 96% Overlap
Restates Superion 256 NVAQC installation, NVQLink, and CUDA-Q integration with zero new facts[cite: 37, 39].

TEST 118
Update — +25% New Info
Discloses SkyWater $1.8B acquisition, Heights Capital $2B funding, and Amazon $36.7M stake[cite: 37, 40].

TEST 119
Conflict — Dispute (128-Qubit)
Contradicts QPU hardware: claims Superion 128 system installed instead of Superion 256[cite: 37, 41].

TEST 120
Different — 0% Overlap
IonQ Superion 256 academic sale & deployment at Florida International University starting Group 048[cite: 37, 42].

Technical Analysis: Dense Vector Blindness vs. Temporal Contradiction Metrics

A core challenge in automated financial news aggregation is that dense vector models (e.g., BAAI/bge-small-en-v1.5, OpenAI Ada-002) compute cosine similarity based on *semantic context* rather than *hardware capacity precision*. Modifying qubit counts (e.g., Superion 256 [256-qubit] in test_116 vs Superion 128 [128-qubit] in test_119) yields extremely high vector similarity (~0.93) because both texts share identical surrounding context tokens ("IonQ", "Superion", "NVIDIA", "NVAQC", "NVQLink", "CUDA-Q").

```
// Formula: Formal Conflict Detection Condition
Conflict_State = ( Cosine_Similarity(E_orig, E_cand) > τ_sim ) && ( ΔF_contradiction == 1.0 )
where τ_sim = 0.60, and ΔF_contradiction = 1.0 indicates a hard hardware specification or qubit capacity dispute.
```

1. Dense Embedding Blindness
Dense embeddings project words into high-dimensional space where "Superion 256" and "Superion 128" share near-identical context embeddings. Thus, pure vector cosine distance fails to flag this hardware spec conflict.

2. LLM Evaluation Override
To prevent corrupting quantum computing product roadmaps, the LLM reasoning layer performs explicit attribute-value model verification, overriding high vector similarity when $\Delta F_{contradiction} = 1.0$ is detected.

Article Relationship & Percentage Breakdown (Batch 024)

FILE IDENTIFIER	PATTERN LABEL	KEYWORDS IDENTIFIED	SIMILARITY %	DELTA %	CLASSIFICATION RATIONALE
`test_116_original`	Original (%5=1)	IonQ Superion 256, NVIDIA NVAQC, NVQLink, GB200 NVL72, CUDA-Q[cite: 37, 38]	100%	Baseline	Establishes breaking story frame on IonQ Superion 256 deployment at NVIDIA NVAQC[cite: 37, 38].
`test_117_repeat`	Repeat (%5=2)	Superion 256 installation, NVAQC 2027, NVQLink, CUDA-Q hybrid research[cite: 37, 39]	96%	0% New Facts	Restates Superion 256 NVAQC installation, NVQLink, and CUDA-Q integration without new developments[cite: 37, 39].
`test_118_update`	Update (%5=3)	**SkyWater $1.8B acquisition**, Heights Capital $2B funding, Amazon $36.7M stake[cite: 37, 40]	81%	+25% Corporate Context	Advances event state via SkyWater $1.8B acquisition details, Heights Capital $2B funding, and Amazon $36.7M stake context[cite: 37, 40].
`test_119_conflict`	Conflict (%5=4)	**Superion 128 dispute**, 128-qubit QPU claim, competing NVAQC account[cite: 37, 41]	93% Match	$\Delta F_{contradiction} = 1.0$	Directly disputes QPU hardware, claiming IonQ is installing a 128-qubit Superion 128 system instead of the Superion 256[cite: 37, 41].
`test_120_different`	Different (%5=0)	Superion 256, Florida International University (FIU), Miami data center, academic sale[cite: 37, 42]	0%	Group 048 Trigger	Zero domain overlap; triggers standalone Group 048 event cluster for IonQ Superion 256 sale to FIU[cite: 37, 42].

Group 047 — Consolidated Event Summary

4 Articles (116–119)

IonQ announced it will deploy an on-premises quantum processing unit (QPU) at NVIDIA's Accelerated Quantum Research Center (NVAQC) in Boston, connecting directly to an NVIDIA GB200 NVL72 system via NVQLink and CUDA-Q for hybrid quantum-classical research in 2027 (**Articles 116–117**)[cite: 37, 38, 39]. Manufacturing foundation and corporate context advanced as **IonQ disclosed that the deployment uses its 6th-gen Superion platform fabricated by SkyWater (acquired by IonQ for $1.8B in Jan 2026), backed by prior funding from Heights Capital ($2B) and Amazon ($36.7M stake)** (**Article 118**)[cite: 37, 40]. However, the quantum processor model is in direct conflict: original reports confirmed deployment of the 256-qubit Superion 256 system, whereas a contradictory report claims **IonQ is installing a 128-qubit Superion 128 system instead** (**Article 119** ; $\Delta F_{contradiction} = 1.0$)[cite: 37, 41].

✓ Update Preserved: SkyWater $1.8B Acquisition, $2B Funding & Amazon Stake
✓ Conflict Preserved: Superion 256 (256-qubit) vs Superion 128 (128-qubit)
✓ Duplicate Removed: Article 117 Consolidated

Group 048 — Standalone Summary

1 Article (120)

IonQ announced an academic partnership contract with Florida International University (FIU) to deploy Florida's first on-premises Superion 256-qubit quantum computer at an on-campus data center in late 2027 (**Article 120**)[cite: 37, 42].

✓ New Event Cluster Established ✓ Zero Entity Contamination from Group 047

Batch 024 Raw Data Source Inspector

JSON Format Verification & Evidence Code Snippets

test_116_original.json **Group 047**


```
{
  "id": 61947635,
  "headline": "IonQ Makes First On-Premise Quantum Dep
  "symbols": ["IONQ", "NVDA", "WQTM", "XITK", "XQTM"],
  "summary": "IonQ shares trade higher after the compa
}
```


Establishes Superion 256 NVAQC deployment baseline.

test_118_update.json **Group 047**


```
{
  "id": 990040018,
  "headline": "IonQ to deploy first quantum computer a
  "symbols": ["IONQ", "NVDA"],
  "summary": "The Superion 256 will be installed in 20
}
```


Adds SkyWater $1.8B deal, $2B funding & Amazon stake.

test_119_conflict.json **Group 047**


```
{
  "id": 990040019,
  "headline": "IonQ To Install Superion 128 As First Q
  "symbols": ["IONQ", "NVDA", "WQTM", "XITK", "XQTM"],
  "summary": "IonQ shares trade higher after the compa
}
```


Disputes QPU spec: Superion 256 vs 128.